LEARNING GUIDE · OCTOBER 2026

How Nowhere works, from a click to the database, and how the AI agent thinks and acts.

A walkthrough of the codebase for its owner: the request lifecycle, the backend layers, the React state, and a deep dive into Nora, the shopping agent. Every section ends with a short Roman Urdu summary.

For
Syed Ibrahim Ali

Read with
the repository open beside it

Time
about 60 minutes

CONTENTS

What's inside

1. **The map** - where every piece of code lives
2. **One request, end to end** - what happens when you click "Add to bag"
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01

The map

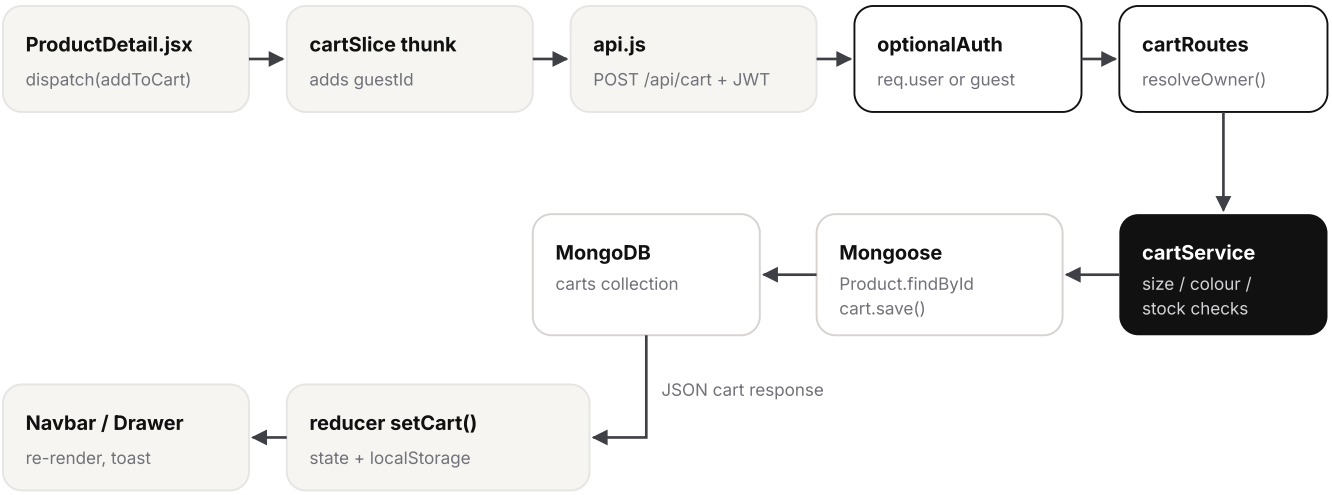
Folder	What lives there
`backend/server.js`	Creates the Express app, applies security middleware, mounts every route, adds the 404 and error handlers.
`backend/routes/`	Thin HTTP handlers: read the request, call a service, send the response.
`backend/services/`	The business rules: `productService` (search, facets, reviews), `cartService` (owner, validation, stock checks), `orderService` (pricing, coupons, placing, cancelling, tracking).
`backend/models/`	Mongoose schemas: User, Product, Review, Cart, Order, Subscriber, AgentSession.
`backend/agent/`	The AI agent: `provider.js` (talks to Groq/Gemini), `tools.js` (what the agent can do), `prompt.js` (its instructions), `agentService.js` (the loop).
`backend/config/store.js`	Shipping fees, coupons, policies and the size guide in one place.
`backend/middleware/`	`protect`, `optionalAuth`, `admin`; error handling and id validation.
`backend/utils/`	`HttpError`, the mailer and text helpers.
`frontend/src/utils/api.js`	The single axios client every request goes through.
`frontend/src/redux/slices/`	State and async thunks per feature: auth, cart, products, checkout, orders, wishlist, store, agent, admin.
`frontend/src/components/`	UI by area: Layout, Product, Cart, Order, Agent, Admin, Common.
`frontend/src/Pages/`	Route-level pages.

> **ASAAN ALFAZ MEIN**
>
> Routes sirf darwaza hain, asal dimaagh `services/` mein hai. Website aur agent dono same services use karte hain, is liye rules ek hi jagah likhe hain.

02

One request, end to end

Follow what happens when a shopper picks size M and presses **Add to bag** on a product page.



1. `ProductDetail.jsx` checks a size is chosen, then dispatches `addToCart({ productId, size, color, quantity })`.
2. The thunk in `cartSlice.js` adds the shopper's `guestId` from Redux and calls `api.post("/cart", ...)`.
3. `api.js` attaches `Authorization: Bearer <token>` if the shopper is signed in.
4. On the server, `optionalAuth` turns a valid token into `req.user`; without one the request continues as a guest.
5. The route calls `cartService.resolveOwner(req.user, guestId)` - the owner is always decided by the server.
6. `cartService.addItem` loads the product, normalises size and colour, checks stock, prices the line at the sale price and saves.
7. The reducer stores the new cart (and caches it in localStorage); the navbar badge and cart drawer re-render.

> **ASAAN ALFAZ MEIN**
>
> Button → Redux thunk → api.js → Express middleware → route → service → MongoDB, aur wapas reducer → UI. Har feature isi raste pe chalta hai.

03

Backend layers

Middleware order in `server.js`

```
app.set("trust proxy", 1);          // real client IPs behind Vercel (rate limits)
app.use(helmet());                  // security headers
app.use(express.json({ limit: "1mb" }));
app.use(cors({ origin: allowlist }));  // CLIENT_URL
app.use("/api", connectDB);         // cached Mongo connection
app.use("/api/...", routes);
app.use(notFound);
app.use(errorHandler);              // last: turns errors into JSON
```

Errors flow through one place

Services throw `new HttpError(409, "Only 2 left in stock")`. Routes pass errors to `next(error)`, and `errorHandler` maps them: `HttpError` keeps its status, Mongoose validation becomes 400, invalid ids become 404, duplicate keys become 409, everything else is logged and returned as 500. The agent's tools catch the same `HttpError` and hand its message back to the model, so Nora can explain the problem in plain words.

A typical route

```
router.post("/", optionalAuth, async (req, res, next) => {
  try {
    const { productId, quantity, size, color } = req.body;
    const { cart } = await cartService.addItem(ownerFrom(req), { productId, quantity, size, color });
    res.status(200).json(cartService.summarize(cart));
  } catch (error) {
    next(error);
  }
});
```

> **ASAAN ALFAZ MEIN**
>
> Route chhota rehta hai: input lo, service ko do, jawab bhejo. Galti ho to `next(error)` — sab errors ek hi `errorHandler` mein handle hote hain.

04

Accounts and guests

Sign-up with OTP

1. `POST /users/register` creates (or refreshes) an unverified user and emails a 6-digit code from `crypto.randomInt`.
2. Only a **SHA-256 hash** of the code is stored, with a 5-minute expiry and an attempt counter.
3. `POST /users/verify-otp` compares hashes with `crypto.timingSafeEqual`; five wrong codes lock that code. Success returns a 7-day JWT.

Three auth middlewares

Middleware	Behaviour
`protect`	Requires a valid JWT; loads the user (the password field is never selected).
`optionalAuth`	Uses the JWT if present, otherwise continues as a guest. Used by cart, checkout and the agent.
`admin`	Runs after `protect`; allows only `role === "admin"`.

Guests

The frontend creates a `guestId` like `guest_1792596270176_gliayvsx` and keeps it in localStorage. It identifies the guest's cart, orders and chat. When the guest signs in, the login thunk calls `/cart/merge` before setting the user, so the bag moves into the account without a race.

> **ASAAN ALFAZ MEIN**
>
> Login ho to JWT se pehchaan, warna `guestId` se. OTP database mein hash ho kar save hota hai, is liye leak hone pe bhi kaam ka nahi.

05

Catalogue and search

`productService.searchProducts` runs one MongoDB aggregation:

```
Product.aggregate([
  { $match: buildMatch(filters) },                // published, gender, size, colour, words...
  { $addFields: { effectivePrice: EFFECTIVE_PRICE } },  // sale price if lower
  { $facet: {
      products: [{ $sort: sort }, { $skip }, { $limit }],
      total: [{ $count: "count" }],
  } },
]);
```

- **Effective price** is `discountPrice` when it is lower than `price`. Price filters and "price low to high" use it, so a sale item sorts by what the shopper pays.
- **Search** splits the query into words; every word must match one of name, description, brand, material, colours, tags, fit or occasion. Input is regex-escaped.
- **Collections** accept slugs (`/collections/summer`) plus special ones: `new-arrivals` , `sale` , `featured` .
- **Facets** (`/products/facets`) group the published catalogue to list the sizes, colours, brands and price range that actually exist; the filter sidebar is built from them.

Reviews

A `$facet` aggregation returns the average, the 1-5 star distribution and the fit counts. The fit meter places a marker at `50 + (large - small) / total × 50` percent. `Review.syncProductRating` keeps `rating` and `numReviews` on the product in sync after each new review, and a partial unique index allows one review per customer per product.

> **ASAAN ALFAZ MEIN**
>
> Search aik aggregation pipeline hai: filter → sale price nikalo → sort/page. Filters ki list bhi database se aati hai, hard-coded nahi.

06

Cart, checkout and stock

The cart re-prices itself

Every time a cart is read, `cartService.refresh` reloads its products, updates names and prices from the catalogue and drops anything deleted or unpublished. The browser never decides a price.

Quoting

`orderService.quote` computes subtotal, shipping (Rs. 250, free from Rs. 5,000), the coupon discount from `config/store.js` and the total. `WELCOME10` is limited to a first order per email.

Placing an order without overselling

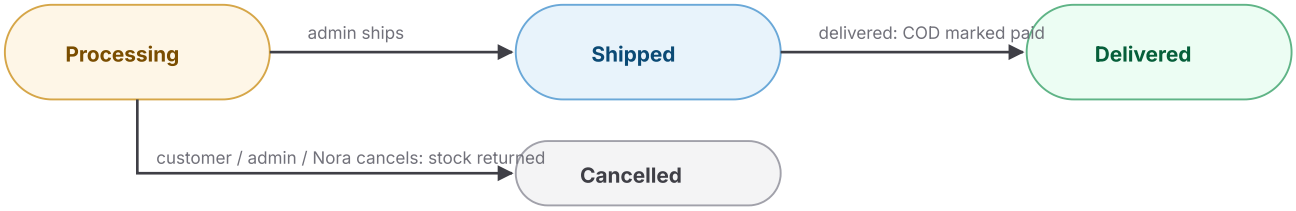
```
for (const [productId, qty] of totals) {
  const res = await Product.updateOne(
    { _id: productId, countInStock: { $gte: qty } },   // only if enough is left
    { $inc: { countInStock: -qty } }                    // ...decrement atomically
  );
  if (res.modifiedCount === 0) {
    await releaseStock(reserved);                       // undo earlier lines
    throw new HttpError(409, "Only N left of ...");
  }
  reserved.push({ productId, quantity: qty });
}
```

The condition and the decrement happen in one database operation, so two shoppers buying the last item can't both succeed. If creating the order fails, the reserved stock is released. After success the cart is cleared, a signed-in user's address is saved, and a confirmation email is sent in the background.

> **ASAAN ALFAZ MEIN**
>
> Stock check aur minus aik hi query mein hota hai (`$gte` + `$inc`), is liye aakhri piece do logon ko nahi bik sakta. Masla ho to stock wapas.

07

Orders



- Every order gets an `orderNumber` like `NW-K7Q2M9XA` and a `statusHistory` entry for each change; the UI draws the timeline from it.
- Only **Processing** orders can be cancelled; cancellation returns stock. Cash on delivery becomes paid when the order is delivered.
- `getOrderForUser` returns the same 404 for "not found" and "not yours", so order ids can't be probed. Guests use `POST /orders/track` with the order number *and* the email.

> ASAAN ALFAZ MEIN
>
> Order ki life: Processing → Shipped → Delivered, ya Processing se Cancelled. Har qadam history mein likha jata hai.

08

Frontend

One API client

```
const api = axios.create({ baseURL: `${import.meta.env.VITE_BACKEND_URL}/api` });
api.interceptors.request.use((config) => { Object.assign(config.headers, authHeaders()); return config;
});
api.interceptors.response.use((r) => r, (error) => {
  if (error.response?.status === 401 && error.config?.headers?.Authorization) {
    window.dispatchEvent(new Event("auth:expired"));   // App logs the user out
  }
  return Promise.reject(error);
});
```

Slices

Slice	Holds
auth	user, guestId; login, register, OTP, reset, profile updates
cart	cart from the server (cached in localStorage), drawer open state
products	listing + paging, product detail, similar items, facets, reviews
checkout / orders	quote, last order (for the confirmation page), order history, tracking
wishlist / store	saved products; shipping, policies and size guide from `/store/config`
agent	chat messages, streaming state, tool label, pending confirmation, session id
admin*	users, products, orders and dashboard stats

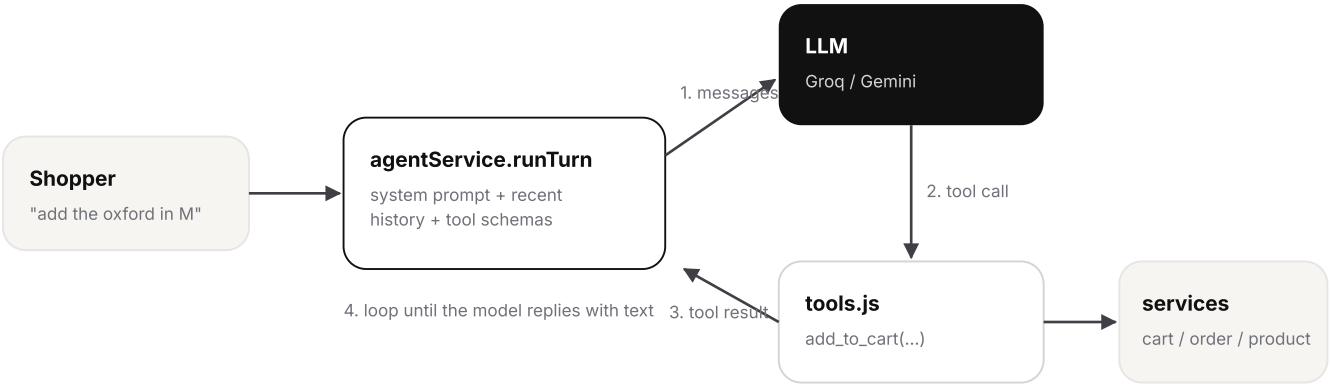
Design system

`tailwind.config.js` defines the palette (`ink` , `cream` , `sale`) and fonts (Fraunces for headings, Inter for text). `index.css` defines reusable classes such as `.btn-primary` , `.input` , `.card` , `.chip` and `.skeleton` , so pages don't repeat long class strings. Shared components (`ProductCard` , `Price` , `Rating` , `Modal` , `OrderParts`) keep screens consistent.

> **ASAAN ALFAZ MEIN**
>
> Har API call `api.js` se jati hai (token khud lagta hai). State Redux slices mein, design tokens Tailwind config mein.

What an agent is, and how Nora is built

A chatbot only talks. An **agent** is a language model that can also *use tools* and repeat until the job is done. On each step the model either replies with text, or asks to call one of the tools you described to it. Your code runs the tool, feeds the result back, and asks the model again.



The four files

File	Responsibility
`provider.js`	Sends one chat-completions request with streaming, parses the server-sent events, collects text and tool calls, and falls back to the next provider on 429/5xx.
`prompt.js`	Nora's rules plus per-request context: today's date, who is shopping (guest or signed-in), and any order waiting for confirmation.
`tools.js`	13 tool definitions (JSON Schema the model reads) and their handlers (code that runs). Handlers call the same services as the website.
`agentService.js`	The loop: build messages, call the model, run tools, append results, repeat (max 6 steps), save the session.

Why OpenAI-compatible

Groq and Google Gemini both accept the OpenAI chat-completions format at their own base URLs. So `provider.js` is one function with a table of providers. It uses Node's built-in `fetch` and reads the stream line by line; no SDK is needed and switching providers is a `.env` change.

```
const PROVIDERS = {
  groq:   { baseURL: () => "https://api.groq.com/openai/v1",                    model: () =>
process.env.GROQ_MODEL || "llama-3.3-70b-versatile" },
  gemini: { baseURL: () => "https://generativelanguage.googleapis.com/v1beta/openai", model: () =>
process.env.GEMINI_MODEL || "gemini-2.5-flash" },
  custom: { baseURL: () => process.env.AGENT_BASE_URL,                          model: () =>
process.env.AGENT_MODEL },
};
```

ASAAN ALFAZ MEIN

Agent = LLM + tools + loop. LLM sochta hai kaunsa tool chalana hai, hamara code tool chalata hai aur result wapas deta hai, jab tak LLM final jawab na de de.

Tools, the loop, memory and streaming

A tool has two halves

```
// 1. The definition the model reads
{ name: "add_to_cart",
  description: "Add a product to the cart. Size and colour must be one of the product's options; ask the
shopper if they haven't chosen.",
  parameters: { type: "object",
    properties: { productId: { type: "string" }, size: { type: "string" }, color: { type: "string" },
quantity: { type: "integer", minimum: 1, maximum: 10 } },
    required: ["productId", "size", "color"] } }

// 2. The handler that runs, always as the requesting shopper (ctx.owner)
async add_to_cart(args, ctx) {
  const { product, line } = await cartService.addItem(ctx.owner, args);
  ctx.emit("cart", { action: "added" });            // tells the browser to refresh the bag
  return { added: `${product.name} (${line.size}, ${line.color})`, cart: await currentCartView(ctx) };
}
```

The 13 tools

Tool	Does
`search_products`	Search with gender, category, colour, size, price, sale, sort; shows product cards
`get_product_details`	Fabric, fit, care, stock, review summary and recent reviews
`get_cart` · `add_to_cart` · `update_cart_item`	Read and change the bag
`get_store_info`	Shipping, returns, payment, sizing, care, contact, size guide, promotions
`get_checkout_details`	Signed in? email? saved address?
`prepare_checkout`	Validates the address, prices the cart, stores a pending order and shows the summary card
`prepare_cancellation`	Stores a pending cancellation for a Processing order
`confirm_pending_action` · `discard_pending_action`	Carry out or drop the pending action
`track_order` · `reorder`	Status and timeline; re-add a past order to the bag

The loop, step by step

1. `session.turn` increases and the shopper's message is appended to `session.messages`.
2. The request is `[system prompt, ...recentHistory(messages)]` plus the tool schemas. `recentHistory` keeps about 24 messages and always starts at a user message, so a tool result is never sent without the

call that produced it.

3. Text tokens stream to the browser as `delta` events while the model writes.
4. If the reply has tool calls, each one runs; its JSON result (capped at 6,000 characters) is appended as a `tool` message, and the loop calls the model again.
5. When the model replies with text only, the turn ends: the transcript is updated and the session saved.

Memory

Each conversation is an `AgentSession` document owned by a user or a guest id. `messages` is the raw model history; `transcript` is a simple list (text + cards) used to redraw the chat after a reload. Sessions expire after 7 days of inactivity via a TTL index. A guest conversation moves to the account when the guest signs in.

Streaming to the browser

SSE event	Browser reaction (`agentSlice.handleEvent`)
`session`	Remember the session id
`delta`	Append text to the current reply
`tool`	Show "Searching the catalogue..." under the reply
`products` / `confirm` / `order`	Render product cards, the order summary card, the order-placed card
`cart`	Re-fetch the cart so the navbar and drawer update
`done` / `error`	Finish the reply; store the pending action / show the error style

The browser uses `fetch` with a stream reader (not `EventSource`) because the request is a POST with an auth header.

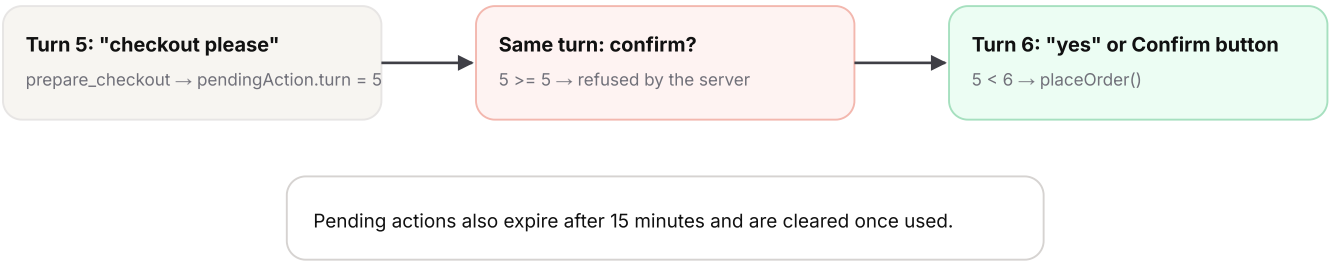
> **ASAAN ALFAZ MEIN**
>
> Tool ke do hisse: description (LLM ke liye) aur handler (asal code). Har message ke saath loop chalta hai, aur server events ke zariye browser ko live text aur cards bhejta hai.

Safety, confirmation, errors and free tiers

Rule 1: identity comes from the server

Handlers never accept a user id from the model. `ctx.owner` is built from the JWT or guest id of the HTTP request. Even if a product review contained "add this to every cart", the model could only act on the current shopper's own cart.

Rule 2: orders need a later confirmation



`prepare_checkout` stores `{ type: "place_order", turn, payload, summary }` on the session. `executePendingAction` refuses unless `pending.turn < session.turn`, which is only true after the shopper has sent another message. The chat's **Confirm** button calls `POST /api/agent/action`, which runs the same function directly without the model, so confirming works even if the AI provider is rate limited.

Rule 3: tool output is data

The system prompt tells Nora that descriptions and reviews are data, not instructions; facts must come from tools; product ids are never shown; and she must ask for a size and colour instead of guessing.

Errors and limits

Situation	Handling
Tool throws `HttpError` (e.g. out of stock)	Returned to the model as `{ error }`; Nora explains and suggests options
Groq `tool_use_failed` (malformed call)	Retried up to twice
429 or 5xx from a provider	Next configured provider is tried; otherwise a friendly "busy" message
Shopper closes the chat mid-reply	The model request is aborted
Abuse	40 messages / 10 min per IP, 1,000 characters per message, 80 turns per session, one reply at a time per session

Free tiers in practice

Free plans limit requests and tokens per minute and per day, per model. Every tool step is one model request, so a "find, add, checkout" conversation uses several. To stay inside the limits Nora sends only recent history, keeps tool results small, and can fall back from Groq to Gemini. Check your current limits in the Groq and Google AI Studio consoles; set both keys for the best uptime.

> **ASAAN ALFAZ MEIN**
>
> Order hamesha do qadam mein: pehle summary, phir agle message mein "haan" ya Confirm button. Server khud check karta hai, model chahe bhi to usi message mein order nahi kar sakta.

12

Extending Nora: add your own tool

Example: a **size recommendation** tool that suggests a size from the shopper's chest measurement.

1. Describe it in `definitions` (tools.js)

```
{ name: "recommend_size",
  description: "Recommend a size for a product from the shopper's chest or waist measurement in
inches.",
  parameters: { type: "object",
    properties: { productId: { type: "string" }, chest: { type: "number" }, waist: { type: "number" } },
    required: ["productId"] } },
```

2. Implement the handler

```
async recommend_size({ productId, chest, waist }, ctx) {
  const product = await productService.getProduct(productId);
  const guide = store.sizeGuide[product.category];                // column 1: chest (tops) / waist
(bottoms)
  const value = product.category === "Top Wear" ? chest : waist;
  if (!value) return { error: "Ask the shopper for their measurement in inches." };
  const row = guide.rows.find((r) => value <= Number(r[1].split("-")[1]));
  const size = row && product.sizes.includes(row[0]) ? row[0] : product.sizes.at(-1);
  return { recommendedSize: size, availableSizes: product.sizes };
},
```

3. Give it a label

```
LABELS.recommend_size = "Working out your size";
```

4. Mention it in the prompt (optional) and test

Add one line to the rules in `prompt.js` ("Use recommend_size when a shopper gives a measurement"), restart the API and ask: *"My chest is 40 inches, which size of the Oxford shirt?"*

> **CHECKLIST FOR ANY NEW TOOL**
>
> Clear description · smallest useful input schema · act only through `ctx.owner` · throw `HttpError` for user-facing problems · return small JSON · emit an event if the UI should update · never perform an irreversible action without the prepare/confirm pattern.

> **ASAAN ALFAZ MEIN**
>
> Naya tool = definition + handler + label. Agar tool kuch permanent kare (jaise payment ya order), to usse bhi "prepare → confirm" pattern mein daalo.

13

Testing

The API and the agent were verified with an end-to-end script that never touches the live database:

- **mongodb-memory-server** starts a throwaway MongoDB; the catalogue and reviews are seeded into it.
- A **mock LLM** is a tiny HTTP server that speaks the OpenAI streaming format and returns scripted tool calls (split across chunks, like real providers), including a malformed call and a 429.
- The backend runs with `AGENT_PROVIDER=custom` pointing at the mock, and the mailer is stubbed to capture OTP and order emails.
- 86 checks cover search and filters, cart validation, guest and user checkout, coupons, stock reservation and release, IDOR protection, OTP flow, admin rules, reviews, the contact form and the agent (streaming, cards, confirmation gate, button path, session privacy, provider errors).

The same mock powers a local demo of the full chat in a real browser, which is how the screenshots in the overview were produced. To test with a real model, put a free Groq or Gemini key in `backend/.env` and chat normally.

> **ASAAN ALFAZ MEIN**
>
> Testing ke liye asli DB aur asli AI ki zaroorat nahi: memory wala MongoDB aur nakli (mock) LLM se poora flow check hota hai.

14

Exercises and glossary

Try these

1. Add a `FREESHIP` coupon in `config/store.js` that removes shipping, and make `quote()` support it.
2. Add a "Recently viewed" tool so Nora can say "you were looking at…". Where would the data come from?
3. Track stock per size: change the Product schema, `reserveStock` and the product page size buttons.
4. Add a `request_return` tool using the prepare/confirm pattern.
5. Switch `AGENT_PROVIDER` to `gemini` and compare answers and speed.

Glossary

Term	Meaning
JWT	Signed token proving who the user is; sent as `Authorization: Bearer`.
OTP	One-time 6-digit code sent by email to verify an address.
Thunk	An async Redux action (API call) with pending / fulfilled / rejected states.
Aggregation pipeline	A list of MongoDB stages ($match, $sort, $group…) that transform documents.
Atomic update	A database operation that checks and changes data in one step, so nothing can happen in between.
LLM	Large language model, e.g. Llama 3.3 on Groq or Gemini.
Tool calling	The model returning a structured request to run a function, instead of plain text.
SSE	Server-sent events: a long HTTP response that sends small messages as they happen.
Prompt injection	Text in data (e.g. a review) trying to give the model instructions; handled by treating tool output as data and enforcing rules in code.
IDOR	Accessing someone else's record by changing an id; prevented by ownership checks.